

Bhavika Jata

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AI & Data Science engineer (B.Tech in Artificial Intelligence and Data Science, latest SGPA: 10/10) specialising in production-ready GenAI systems, RAG pipelines, LLM-powered agents, and FAISS-backed vector search. Contributed to an agentic NLP-to-SQL analytics system at Tata Technologies and automated data pipelines at L&T Precision cutting manual effort by 40%. Built and deployed ResearchEngine, a multi-module AI research assistant with semantic document ingestion and mentor-style LLM conversations. Targeting AI/ML engineering and GenAI product roles.

WORK EXPERIENCE

AI Intern

Tata Technologies Ltd., Pune

Dec 2025 – Jan 2026

- Contributed to a tool-mode agentic analytics system on the Langflow framework, where an LLM agent delegated query execution to a custom tool, enabling HR staff to query datasets in plain English.
- Implemented Fernet-based encryption for sensitive data fields (gender, age, income) with controlled in-memory decryption, per project security requirements.
- Extended the system's ingestion pipeline for multi-sheet Excel and CSV files, automatically converting uploads into structured, queryable SQLite tables.

Data Analyst Intern

L&T Precision Engineering and Systems, Mumbai

May 2025 – Jun 2025

- Automated Python-based web scraping (Selenium, BeautifulSoup) across defence and general news sites, collecting 429 relevant articles mentioning L&T and its products.
- Built an SVM-based sentiment analysis model (TF-IDF vectorized) attaining 78.0% accuracy classifying media mentions as positive, neutral, or negative.
- Delivered an interactive Power BI dashboard visualizing sentiment distribution and product-level mention trends for strategy teams to monitor media perception.

EDUCATION

K. J. Somaiya School of Engineering, Mumbai

B.Tech in Artificial Intelligence & Data Science (Honors in Data Engineering)

Aug 2023 – May 2027

SGPA: 10/10 | CGPA: 8.908/10

Relevant Coursework: Data Structures & Algorithms, Machine Learning, Deep Learning, Artificial Intelligence, Database Management Systems, Probability & Statistics

PROJECTS

ResearchEngine

May 2026 (ongoing)

- Developed an AI-powered RAG-based LLM research assistant using Python, LangChain, FAISS, and a multi-provider LLM routing architecture (Groq, Gemini), deployed via FastAPI on HuggingFace Spaces with a Netlify frontend for live use.
- Built a multi-format document ingestion and semantic search system for research documents.
- Designed for researchers at all levels from literature discovery and paper analysis to mentor-style AI conversations, benchmarked at 1.8s avg response latency, significantly reducing multi-source search effort.

CNN vs. Quantum CNN — Sugarcane Leaf Disease Detection

Jan 2026 (ongoing)

- Benchmarked Custom CNN, EfficientNet-B0, and MobileNet-V2 transfer learning architectures trained on a 13-class sugarcane disease dataset (12,000+ images), evaluating generalization on a held-out 5-class, 2,521-sample test set.
- EfficientNet-B0 achieved the highest test accuracy of 96.79% (precision, recall, F1-score: 0.97), outperforming Custom CNN (84.73%) and MobileNet-V2 (82.43%) on the test benchmark.
- Extending the study toward Quantum CNN architectures to evaluate whether quantum feature extraction can surpass the classical benchmark, as part of a classical vs. quantum comparative analysis.

Crop Recommendation & Yield Prediction System

Nov 2023

- Engineered ensemble ML models (Random Forest, XGBoost, Voting Classifier) in Python achieving 99.52% prediction accuracy on a scalable pipeline processing 10,000+ records across 21 crop types.
- Integrated MODIS satellite imagery via Google Earth Engine for geospatial temperature analysis.

Resume Eligibility Parser

Jun 2024

Designed a rule-based NLP engine in Python processing 50+ resumes, eliminating manual review for 80% of unqualified applicants; deployed via Streamlit, cutting decision time from 5 minutes to under 30 seconds.

Social Media Sentiment Analysis

Apr 2023

Built an end-to-end NLP pipeline using SVM (Python/Scikit-learn) achieving 92% accuracy classifying 2,000+ Google Form survey responses on social media usage, cutting collection and cleaning time by 60%.

SKILLS

- Programming:** Python, SQLite, PostgreSQL, MS Excel
- Computer Vision:** Image Classification, Image Processing (OpenCV, Albumentations), CNNs, RNNs, LNNs Transfer Learning (EfficientNet, MobileNet-V2, ResNet), Data Augmentation
- ML / DL Frameworks:** PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, SVM, Random Forest, CNN, RNN, Transformer models
- NLP & AI Orchestration:** LangChain, Langflow, NLP Pipelines, BeautifulSoup, Selenium, RestAPI, RAG pipelines, Vector Databases (FAISS, ChromaDB), Large Language Models (LLMs)
- Version Control:** Git, GitHub
- Data Visualization & BI:** Power BI, Tableau, Matplotlib, Seaborn, Streamlit
- Big Data & Engineering:** Hadoop, PySpark, ETL Pipelines
- Geospatial & Remote Sensing:** Google Earth Engine, QGIS

CERTIFICATIONS

- 10th Summer School on AI, CVIT — IIIT Hyderabad — Jul 2026
Selective program covering Deep Learning, LLMs, VLMs, Diffusion Models, LoRA/PEFT, Supervised Fine-Tuning, LLM Alignment, and RLHF
- Tata — GenAI Powered Data Analytics Job Simulation, Forage — Jun 2026
- Deloitte Australia — Data Analytics Job Simulation, Forage — Nov 2025
- EDA & Data Visualization in Data Science —Scaler, Mar 2025